Form Models

You can use **Pydantic models** to declare **form fields** in FastAPI.

Info

To use forms, first install `python-multipart`.

Make sure you create a virtual environment, activate it, and then install it, for example:

```
$ pip install python-multipart
```

Note

This is supported since FastAPI version `0.113.0`. 🤓

Pydantic Models for Forms

You just need to declare a **Pydantic model** with the fields you want to receive as **form fields**, and then declare the parameter as `Form`:

Python 3.10+

```python
from typing import Annotated

from fastapi import FastAPI, Form
from pydantic import BaseModel

app = FastAPI()


class FormData(BaseModel):
    username: str
    password: str


@app.post("/login/")
async def login(data: Annotated[FormData, Form()]):
    return data
```

Ask AI

🤓 **Other versions and variants**

Python 3.10+ - non-Annotated

Tip

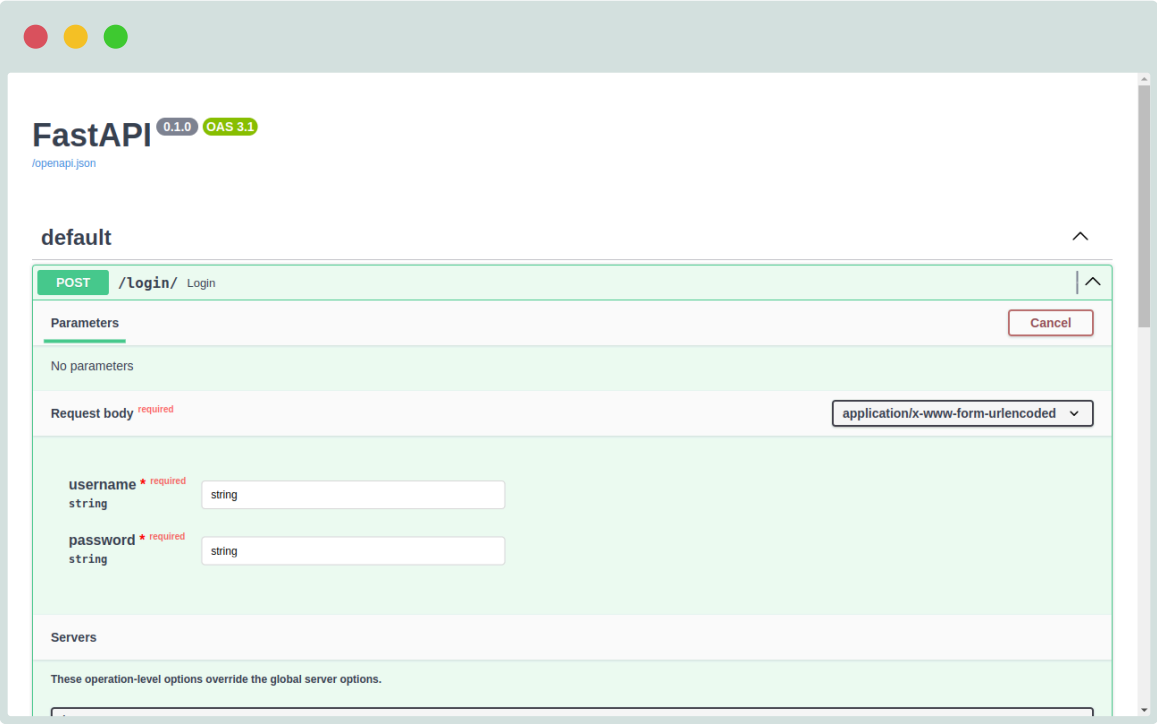
Prefer to use the `Annotated` version if possible.

```python
from fastapi import FastAPI, Form
from pydantic import BaseModel

app = FastAPI()


class FormData(BaseModel):
    username: str
    password: str


@app.post("/login/")
async def login(data: FormData = Form()):
    return data
```

FastAPI will **extract** the data for **each field** from the **form data** in the request and give you the Pydantic model you defined.

Check the Docs

You can verify it in the docs UI at `/docs`:



Forbid Extra Form Fields

In some special use cases (probably not very common), you might want to **restrict** the form fields to only those declared in the Pydantic model. And **forbid** any **extra** fields.

> **Note**
>
> This is supported since FastAPI version `0.114.0`. 🤓

You can use Pydantic's model configuration to `forbid` any `extra` fields:

Python 3.10+

```python
from typing import Annotated

from fastapi import FastAPI, Form
from pydantic import BaseModel

app = FastAPI()


class FormData(BaseModel):
    username: str
    password: str
    model_config = {"extra": "forbid"}


@app.post("/login/")
async def login(data: Annotated[FormData, Form()]):
    return data
```

> **🤓 Other versions and variants**

Python 3.10+ - non-Annotated

> **Tip**
>
> Prefer to use the `Annotated` version if possible.

```python
from fastapi import FastAPI, Form
from pydantic import BaseModel

app = FastAPI()


class FormData(BaseModel):
    username: str
    password: str
    model_config = {"extra": "forbid"}


@app.post("/login/")
async def login(data: FormData = Form()):
    return data
```

If a client tries to send some extra data, they will receive an **error** response.

For example, if the client tries to send the form fields:

- `username`: `Rick`

- `password`: `Portal Gun`
- `extra`: `Mr. Poopybutthole`

They will receive an error response telling them that the field `extra` is not allowed:

```
{
    "detail": [
        {
            "type": "extra_forbidden",
            "loc": ["body", "extra"],
            "msg": "Extra inputs are not permitted",
            "input": "Mr. Poopybutthole"
        }
    ]
}
```

Summary

You can use Pydantic models to declare form fields in FastAPI. 😎